

Task 2 – Product & Category Management

Database: ProductCategoryDB

Objective: Create Category and Product tables, apply constraints, insert sample data, perform CRUD operations, and analyse products category-wise.

1. Category & Product Table Design

Category

Column	Data Type	Constraint
Category_ID	INT	Primary Key
Category_Name	VARCHAR(100)	NOT NULL, UNIQUE
Description	VARCHAR(255)	NOT NULL

Product

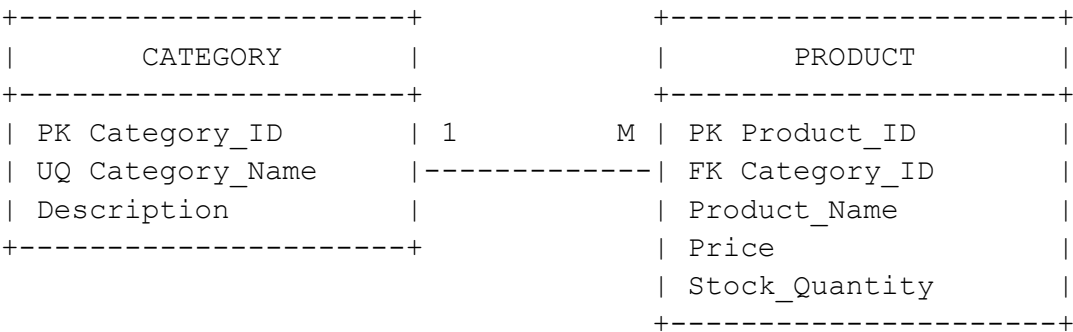
Column	Data Type	Constraint
Product_ID	INT	Primary Key
Product_Name	VARCHAR(100)	NOT NULL, UNIQUE
Category_ID	INT	NOT NULL, Foreign Key
Price	DECIMAL(10,2)	NOT NULL, CHECK > 0
Stock_Quantity	INT	NOT NULL, CHECK >= 0

2. Constraints Applied

- PRIMARY KEY uniquely identifies each category/product record.
- FOREIGN KEY connects Product.Category_ID to Category.Category_ID.
- NOT NULL prevents required fields from being left empty.
- UNIQUE prevents duplicate Category_Name and Product_Name values.
- CHECK ensures Price is greater than 0 and Stock_Quantity is not negative.

3. ER Diagram

One Category can contain many Products. Each Product belongs to one Category.



4. Sample Product Data

Product ID	Product Name	Category ID	Price	Stock
101	Wireless Mouse	1	799.0	50
102	Mechanical Keyboard	1	2499.0	30
103	USB-C Charger	1	1299.0	45
104	Bluetooth Speaker	1	1999.0	25
105	Webcam	1	1599.0	20
106	Power Bank	1	999.0	40
107	Electric Kettle	2	1799.0	15
108	Mixer Grinder	2	3499.0	10
109	Table Fan	2	2299.0	18
110	Notebook	3	120.0	100
111	Ball Pen Pack	3	80.0	150
112	A4 Paper Pack	3	350.0	60

5. Category-wise Analysis Report

The sample data contains 12 products across 3 categories.

Category	Product Count	Highest Price	Average Price
Electronics	6	3499.00	583.17
Home Appliances	3	3499.00	2529.00
Stationery	3	350.00	183.33

Highest-priced product: Mixer Grinder – ₹3,499.00.

Category having more than 5 products: Electronics – 6 products.

Overall average product price: ₹1,421.83.

6. CRUD Operations

INSERT: Added HDMI Cable as Product_ID 113.

SELECT: Displayed all records from Product.

UPDATE: Changed HDMI Cable price to ₹699 and stock to 40.

DELETE: Removed HDMI Cable after demonstrating the DELETE operation.

7. Query Output Screenshots

Run the SQL script in MySQL Workbench/phpMyAdmin and capture screenshots of the result grids for the CRUD and category-wise analysis queries. Insert those screenshots in this section before submission.

8. Conclusion

The database successfully demonstrates table creation, relational constraints, sample product management, CRUD operations, and category-wise analysis using SQL aggregate functions, JOIN, GROUP BY, HAVING, MAX, and AVG.